

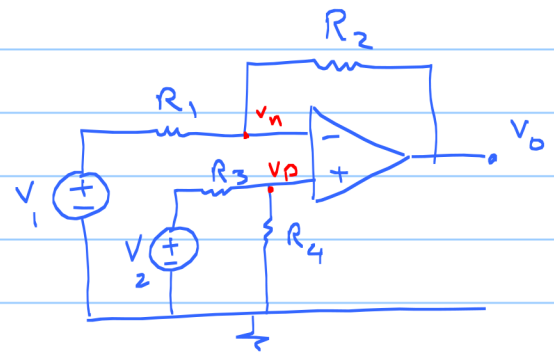


Difference amplifier

$$\frac{V_n - V_1}{R_1} + \frac{V_n - V_o}{R_2} = 0$$

$$V_o = \left(\frac{R_2}{R_1} + 1\right) V_n - \frac{R_2}{R_1} V_1 \quad (1)$$

$$V_n = V_p = V_2 \frac{R_4}{R_3 + R_4} \quad (2)$$



substitute V_n from (2) into (1) :

$$V_o = \left(\frac{R_2}{R_1} + 1\right) \left(\frac{R_4}{R_3 + R_4} V_2\right) - \frac{R_2}{R_1} V_1$$

$$V_o = \frac{R_4}{R_1} \left(\frac{R_1 + R_2}{R_3 + R_4}\right) V_2 - \frac{R_2}{R_1} V_1$$

Divide by $\frac{R_4}{R_1}$, then factor R_2 on the numerator.

$$V_o = \frac{R_1 + R_2}{R_1 \left(1 + \frac{R_3}{R_4}\right)} V_2 - \frac{R_2}{R_1} V_1$$

$$V_o = \frac{R_2 \left(1 + \frac{R_1}{R_2}\right)}{R_1 \left(1 + \frac{R_3}{R_4}\right)} V_2 - \frac{R_2}{R_1} V_1$$

To make a subtractor, the coefficients of V_1 and V_2 must be equal.

$$\frac{R_2 \left(1 + \frac{R_1}{R_2}\right)}{R_1 \left(1 + \frac{R_3}{R_4}\right)} = \frac{R_2}{R_1} \Rightarrow \frac{R_1}{R_2} = \frac{R_3}{R_4}$$

$$\text{Then, } V_o = \frac{R_2}{R_1} (V_2 - V_1)$$

Ex. Rework the previous problem using a difference amplifier.

$$V_o = 3V_a - 2V_b$$

The output voltage for a difference amplifier is :

$$V_o = \frac{R_2 \left(1 + \frac{R_1}{R_2}\right)}{R_1 \left(1 + \frac{R_3}{R_4}\right)} V_2 - \frac{R_2}{R_1} V_1$$

Comparing the expression for V_o in a difference amplifier with the desired V_o :

$$\frac{R_2}{R_1} = 2 \quad \text{and}$$

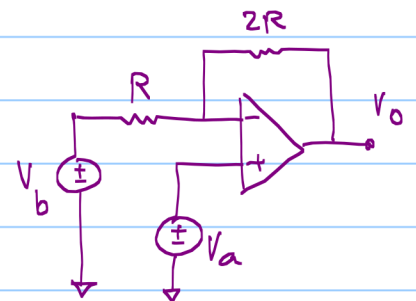
$$2 \frac{R_2 \left(1 + \frac{R_1}{R_2}\right)}{R_1 \left(1 + \frac{R_3}{R_4}\right)} = 3 \Rightarrow \frac{2 \left(1 + \frac{1}{2}\right)}{1 + \frac{R_3}{R_4}} = 3 \rightarrow 1 + \frac{R_3}{R_4} = 1$$

$\Rightarrow R_3 = 0, R_4$ (no effect)

For example:

$$R = 10 \text{ k}$$

$$R_F = 2R = 20 \text{ k}$$



you can check your work using superposition :

$$\text{Turn off } V_a : V_o' = -\frac{2R}{R} V_b = -2V_b \quad (\text{inverting})$$

$$\text{Turn off } V_b : V_o'' = \left(1 + \frac{2R}{R}\right) V_a = 3V_a \quad (\text{non-inverting})$$

$$\Rightarrow V_o = V_o' + V_o'' = \underline{3V_a - 2V_b} \quad \checkmark$$

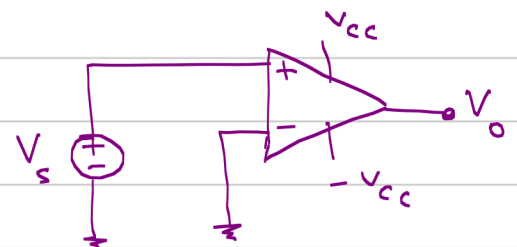
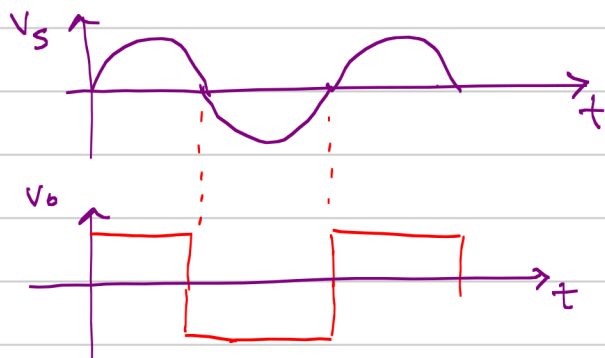
Comparator

A negative feed back keeps the opamp in the linear mode with $V_p \approx V_n$. Without a negative feedback, with a few millivolts between V_p and V_n the opamp becomes saturated.

if $V_p > V_n \Rightarrow V_o = +V_{cc}$ (positive saturation)

if $V_p < V_n \Rightarrow V_o = -V_{cc}$ (negative saturation)

Therefore, the sign of the output indicates which input is larger. An opamp with no feedback connection is used as a comparator.



zero crossing detector

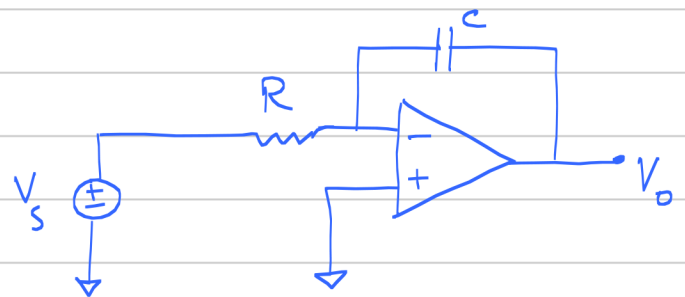
Integrating amplifier circuit

If the feedback resistor in the inverting amplifier circuit is replaced by a capacitor, the output voltage will be proportional to the integral of the input signal. This circuit is called an integrator and used in numerous applications.

$$V_n = V_p = 0$$

$$\frac{0 - V_s}{R} + C \frac{d}{dt} (0 - V_o) = 0$$

$$V_o = -\frac{1}{RC} \int_0^t V_s dt$$



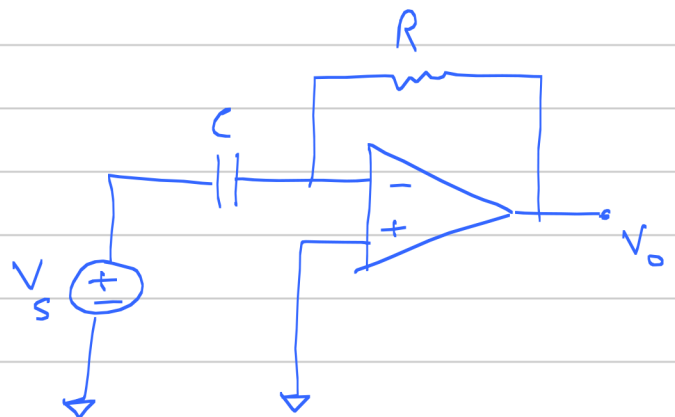
Differentiator circuit

If the input resistor in the inverting amplifier circuit is replaced by a capacitor, the output voltage will be proportional to the differential of the input signal. This circuit is called a differentiator.

$$V_n = V_p = 0$$

$$C \frac{d}{dt} (0 - V_s) + \frac{0 - V_o}{R} = 0$$

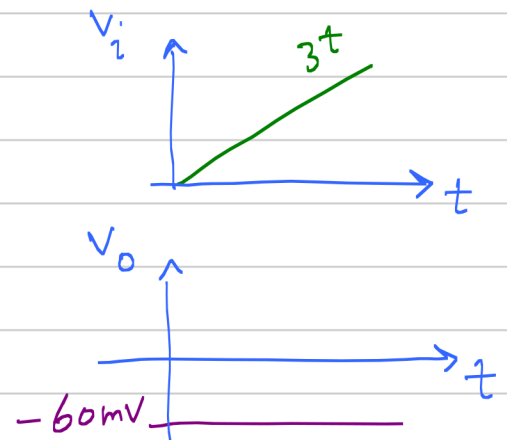
$$V_o = -RC \frac{dV_s}{dt}$$



Ex. A differentiator has $R = 10\text{ k}$ and $C = 2\text{ MF}$.

If $V_i = 3t\text{ V}$, find the output voltage V_o .

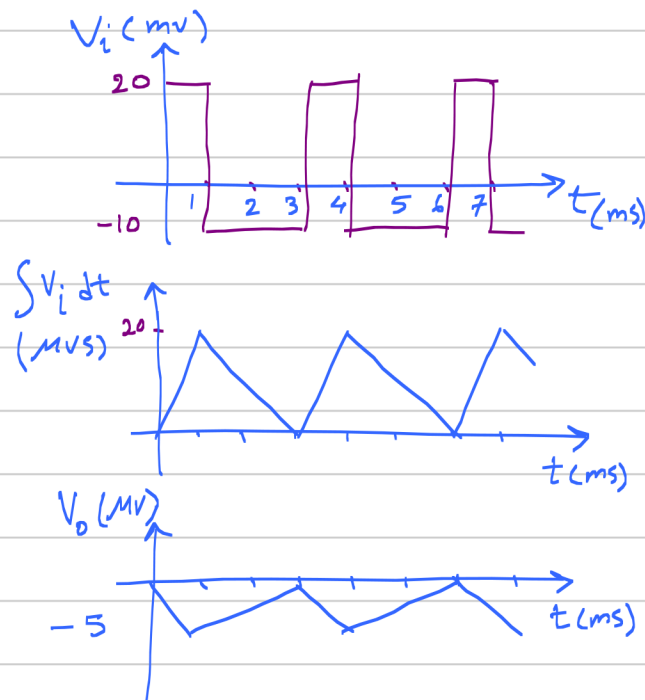
$$V_o = -RC \frac{dV_i}{dt} = -(10^4)(2 \times 10^{-6})(3) = -60\text{ mV}$$



Ex. An integrator with $R = 4\text{ M}\Omega$ and $C = 1\text{ MF}$ has the input waveform shown below. Plot the output waveform.

$$V_o = -\frac{1}{RC} \int_0^t V_i dt$$

$$= -\frac{1}{4} \int_0^t V_i dt$$



Ex. The displacement of a robot arm in a given direction is represented by a voltage signal $v_{in}(t)$. The voltage is proportional to displacement and 1V corresponds to a displacement of 10mm from the reference position.

Design a circuit that produces a voltage $v_1(t)$ that is proportional to the velocity of the robot arm such that 1m/s corresponds to 1V.

Design an additional circuit that produces a voltage v_2 that is proportional to acceleration such that 1m/s² corresponds to 1V.

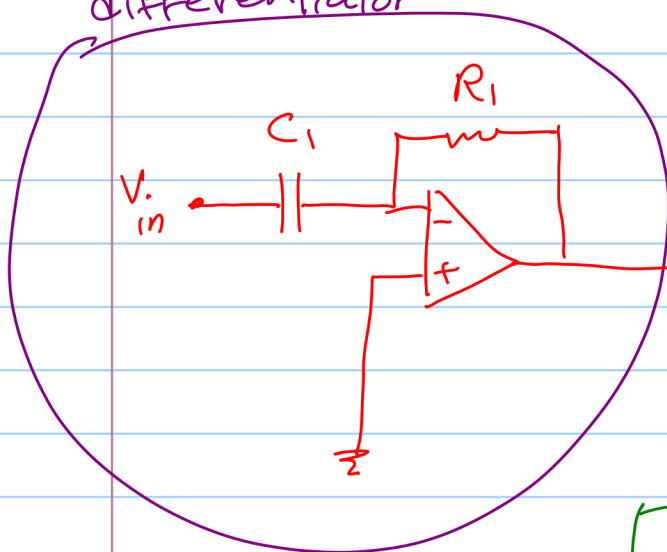
$x(t)$ = displacement in meter

$$\Rightarrow v_{in}(t) = 100 x(t)$$

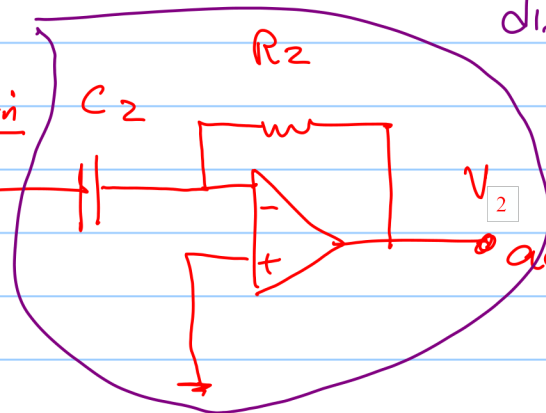
Then velocity $v_1(t) = \frac{dx}{dt} = 0.01 \frac{dv_{in}(t)}{dt}$

and acceleration $v_2(t) = \frac{dx}{dt^2} = \frac{dv_1(t)}{dt}$

differentiator

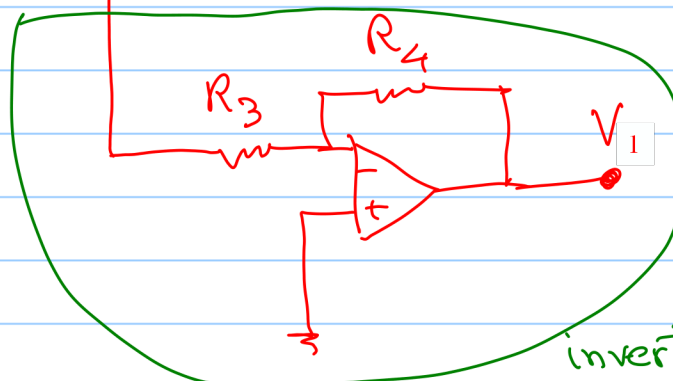


differentiator



requirements

$$\begin{cases} R_1 C_1 = 0.01 \\ R_2 C_2 = 1 \\ R_3 = R_4 \end{cases}$$



velocity

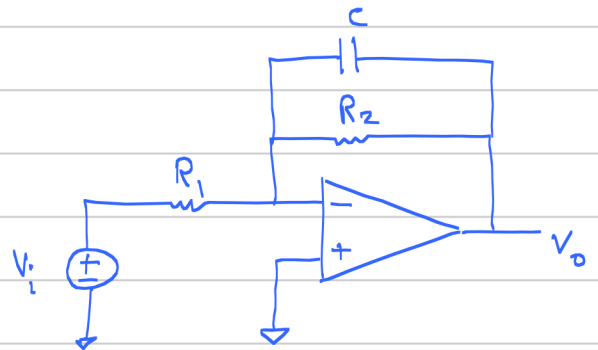
inverting

Active filters

We can design a low pass, high pass, band pass and band reject filters using opamps, resistors and capacitors. Active filters have several advantages over passive filters. They are capable of amplifying the signal as well as filtering. The amplification can be controlled.

Low pass active filter

$$H(f) = \frac{V_o}{V_i} = - \frac{Z_f}{Z_i}$$
$$= - \frac{R_2 \parallel \frac{1}{j2\pi f C}}{R_1}$$



$$= - \frac{\frac{R_2 \cdot \frac{1}{j2\pi f C}}{R_2 + \frac{1}{j2\pi f C}}}{R_1} = - \frac{\frac{R_2}{1 + j2\pi f R_2 C}}{R_1} = - \frac{R_2}{R_1 (1 + j2\pi f R_2 C)}$$
$$= - \frac{R_2}{R_1} \cdot \frac{1}{1 + j \frac{f}{f_B}}, \quad f_B = \frac{1}{2\pi R_2 C}$$

This is the transfer function of a low pass RC circuit, multiplied by a gain of $\frac{R_2}{R_1}$.

High pass active filter

$$H(f) = \frac{V_o}{V_i} = \frac{-Z_f}{Z_i} = -\frac{R_2}{R_1 + \frac{1}{j2\pi fC}}$$

$$= -\frac{R_2}{R_1} \frac{1}{1 + \frac{1}{j2\pi fR_1C}} = -\frac{R_2}{R_1} \frac{1}{1 - j\frac{f_B}{f}}$$

$$= -\frac{R_2}{R_1} \frac{jf}{f_B + jf} = -\frac{R_2}{R_1} \frac{j\frac{f}{f_B}}{1 + j\frac{f}{f_B}}, \quad f_B = \frac{1}{2\pi R_1 C}$$

